Project Design Phase
Solution Architecture

Date	06-April-2026
Team ID	SWTID-2026-2633
Project Name	Automating data import and relationship mapping import sets and Dot walking
Maximum Marks	4 Marks

Solution Architecture:

Components Explanation

1. External Data Source

- Data comes from files (CSV, Excel), REST APIs, or databases
- Contains raw, unstructured or semi-structured data

2. Data Source (ServiceNow)

- Defines how data is imported (file upload, scheduled import, API)
- Acts as a connector between external data and ServiceNow

3. Import Set Table (Staging Layer)

- Temporary table where raw data is stored
- Keeps original data safe before processing
- Allows validation and preprocessing

4. Transform Map (Processing Layer)

- Maps fields from Import Set → Target Table
- Applies:
  - Field mapping
  - Coalesce (duplicate handling)
  - Scripts (onBefore, onAfter)

5. Target Tables (Data Storage Layer)

- Final destination (e.g., Users, Departments, CMDB tables)
- Clean and structured data is stored here

6. Relationship Mapping Layer

- Establishes links between records:
  - Reference fields (User → Department)
  - CI relationships (Parent → Child)
- Implemented using:
  - Transform scripts
  - Lookup rules

Optional Enhancements

- 🔄 **Scheduled Imports** → automatic periodic data loads
- 🔗 **API Integration** → real-time data sync

- 🧹 **Data Validation Rules** → ensure quality
- 📊 **Logging & Error Handling** → track failures

Key Architecture Idea

Staging → Transformation → Storage → Relationship Linking

Outcome

- Automated data import
- No duplicates
- Accurate relationship mapping
- Scalable and reusable system